TOPIC NAME : Lecture 1 (class 1 - online) DAY : 10.02.26

TIME : DATE : / /

$f \propto \frac{1}{\lambda}$

Software : EST → Computer simulation Technology (2020+)

Microwave freq. → form of electromagnetic radition with wave-length ranging 30 cm – 1 mm (1 - 300 GHz)

→ are up to infrared and visible light reasons.

→ We'll use [1 - 10^6 GHz]

→ electromagnetic signal + shorter wavelength

→ not reflected by Ionosphere. ← energy loss for travel

→ straight line but attenuated by short distance

Adv → intallation costing high → interferdence

→ takes space more → dependency on temp.

→ inherent inefficiency of electric power.

Application

→ blutooth, broad cast, military radar system, industrial, medical...

TOPIC NAME : Lecture 2 DAY : 17.02.2026.

TIME : DATE : / /

Electric & magnetic goes perpendicular cause electro-magnetic field.

Electron motion in a Electric field calculation

ϵ_0 = permitivity in air medium

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2}$$

$$= \frac{Q^2}{4\pi\epsilon_0 R^2} \quad \left[\text{Let } Q_1 = Q_2 = Q\right] \text{ newton}$$

We know, electric field, (force per unit charge)

$$E = \frac{F}{Q} = \frac{\frac{Q^2}{4\pi\epsilon_0 R^2}}{Q} = \frac{Q}{4\pi\epsilon_0 R^2} \quad Vm^{-1}$$

For n charge, $E = \sum_{m=1}^{n} \frac{Q_m}{4\pi\epsilon_0 R_m^2}$

$U_r = \cos\phi\, U_x + \sin\phi\, U_y$

$U_\theta = \sin\theta\, U_x + \cos\theta\, U_y$

$U_r = \cos\phi\, U_x + \sin\phi\, U_y$

$$\Rightarrow \frac{d}{d\phi} U_r = \frac{d}{d\phi}\cos\phi\, U_x + \frac{d}{d\phi}\sin\phi\, U_y$$

$$= -U_x \sin\phi + U_y \cos\phi$$

$$= -\sin\phi\, U_x + \cos\phi\, U_y$$

$$= U_\theta$$

$$\frac{d}{d\phi} U_\theta = \frac{d}{d\phi}(-\sin\phi\, U_x) + \frac{d}{d\phi}(\cos\phi\, U_y)$$

$$= -\cos\phi\, U_x - \sin\phi\, U_y$$

$$= -(\cos\phi\, U_x + \sin\phi\, U_y)$$

$$= -U_r$$

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RTT → Round Trip time

Pointing vector, $\rho_p = r u_r + z u_z$

$$\Rightarrow \frac{d\rho}{dt} = \frac{d}{dt}(r u_r) + \frac{d}{dt}(z u_z)$$

$$\Rightarrow \frac{d\rho}{dt} = \frac{dr}{dt} u_r + \frac{du_r}{dt} r + \frac{dz}{dt} u_z + \frac{du_z}{dt} z$$

$$= \frac{dr}{dt} u_r + \frac{du_r}{dt} r + \frac{du_z}{dt} z \quad \left[z \text{ is a constant. so, } \frac{du_z}{dt} = 0 \right]$$

$$= u_r \frac{dr}{dt} + r \frac{du_r}{dt} + u_z \frac{dz}{dt}$$

Lecture 3 (18-02-2026)

Helmohltz equation

Maxwell equation,

$$\nabla \times \bar{E} = -j\omega\mu\bar{H} \quad \text{---} ①$$

$$\nabla \times \bar{H} = j\omega\epsilon\bar{E} \quad \text{---} ②$$

H → magnetic field
E → electric "

From eq. ②,

$$\bar{H} = \frac{j\omega\epsilon\bar{E}}{\nabla}$$

Then, ① be,

$$\nabla \times \bar{E} = -j\omega\mu \frac{j\omega\epsilon\bar{E}}{\nabla}$$

$$\Rightarrow \nabla \times \nabla \times \bar{E} = \omega^2 \mu\epsilon\bar{E} \quad \text{---} (iii)$$

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Vector identity law,

$$\nabla \times \nabla \times \bar{A} = \nabla(\nabla \cdot A) - \nabla^2 \bar{A}$$

(III) $\rightarrow$ $\nabla \times \nabla \times \bar{E} = \nabla(\nabla \cdot E) - \nabla^2 \bar{E}$

$\Rightarrow$ $\omega^2 \mu \epsilon \bar{E} = \nabla(\nabla \cdot E) - \nabla^2 \bar{E}$

$\Rightarrow$ $\nabla \cdot (\nabla \cdot E) = \omega^2 \mu \epsilon E + \nabla^2 \bar{E}$ ——— (IV)

For source free region, $\nabla \cdot E = 0$

বাইরে কোন no power supply

$$\nabla^2 \cdot \bar{E} + \omega^2 \mu \epsilon \bar{E} = 0$$

$\Rightarrow \nabla(\nabla \cdot \bar{E}) = \omega^2 \mu \epsilon \bar{E} + \nabla^2 \bar{E}$ $\rightarrow$ electric field

Similarly. $\nabla^2 \bar{H} + \omega^2 \mu \epsilon \bar{H} = 0$ $\rightarrow$ magnetic field

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Pointing theorem (To much charge happens for changing EF & MF)

Pointing vector for steady state is -

$$\langle P \rangle = \langle E * H \rangle = \frac{1}{2} R_e (E * H^*) \quad \text{(complex conjugate)} \quad \ldots\ldots (1)$$

$$P = \frac{1}{2} (E * H^*) \ldots\ldots (2)$$

From Maxwell's equation,

$$\nabla \times E = -j\omega \mu H \quad \text{——} (3)$$

$$\nabla \times H = j + j\omega \epsilon E \quad \text{——} (4)$$

Dot multiplication of (3) by H^*,

$$(\nabla \times E) H^* = -j\omega \mu H \cdot H^* \quad \text{——} (5)$$

complex conjugate of equation (4), → অর্থাৎ EF এর sign change হবে

$$\nabla \times H^* = J^* - J\omega \epsilon E^* \quad \text{——} (6)$$

(6) · E ⟹ $(\nabla \times H^*) E = (J^* - J\omega \epsilon E^*) E \quad \text{——} (7)$

(7) − (5) ⇒ $E(\nabla \times H^*) - H^*(\nabla \times E) = E \cdot J^* - J\omega \epsilon E^* E + j\omega \mu H^* H$

$[E \cdot E^* = |E|^2]$

$$= E \cdot j^* - j\omega (\epsilon \cdot E \cdot E^* - \mu H^* H)$$

$$= E \cdot j^* - j\omega (\epsilon |E^2| - \mu |H^2|)$$

$$\therefore E(\nabla \times H^*) - H^*(\nabla \times E) = E \cdot j^* - j\omega (\epsilon |E|^2 - \mu |H^2|)$$

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TOPIC NAME : _____ DAY : _____

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$$\nabla \times H^* = J^* - J\omega\epsilon E^* \quad \text{——} (6)$$

$$(6) \cdot E \Rightarrow (\nabla \times H^*) E = (J^* - J\omega\epsilon E^*) E \quad \text{——} (7)$$

$$(7) - (5) \Rightarrow E(\nabla \times H^*) - H^*(\nabla \times E) = E \cdot J^* - J\omega\epsilon E^* E + j\omega\mu H^* H$$

$[E \cdot E^* = |E^2|]$

$$= E \cdot j^* - j\omega (\epsilon \cdot E \cdot E^* - \mu H^* H)$$

$$= E \cdot j^* - j\omega (\epsilon |E^2| - \mu |H^2|)$$

$$\therefore E(\nabla \times H^*) - H^*(\nabla \times E) = E \cdot j^* - j\omega (\epsilon |E|^2 - \mu |H^2|)$$

electromagnetic plane এ কোনো বস্তুর মধ্য দিয়ে তড়িৎ চৌম্বক শক্তি পরিবর্তন হয়

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Radiating region → covers the area and region that network has. [Antenna] → types

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TOPIC NAME : Lecture 6

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Electromagnetic wave propagate in free space in two ways

① Ground wave → Direct wave * — *
→ surface wave → Reflective wave

Plane wave – EM wave

② Sky wave / ionosphere wave → Ionised by solar radiation [x microwave]

- If critical frequency = F_{cr}
  maximum electron density = N_{max}

  $F_{cr} = 9\sqrt{N_{max}}$ e/m³

  ↳ Reflected back to earth from sky wave

Power density and received power

If, P_t = transmitting power in watts

g_t = transmitting antenna gain (Numerical)

R = distance between antenna and field point in meters

The power density,

$$P_d = \frac{P_t g_t}{4\pi R^2} \text{ W m}^{-2} \quad \text{---(1)}$$

The receiving antenna with gain g_r has effective aperture

$A_e = \frac{\lambda^2}{4\pi} g_r$

$\lambda^2/4\pi \rightarrow A_a$ = isotropic antenna aparture (m²)

g_r → receiving antenna gain

So the received power will be,

$$P_r = P_d A_e = \left(\frac{P_t g_t}{4\pi R^2}\right)\left(\frac{\lambda^2}{4\pi} g_r\right) \text{ watts} \quad \text{---(11)}$$

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Converting (2) in decibles,

$$P_r = \left(\frac{g_t P_t}{4\pi R^2}\right)\left(\frac{\lambda^2 g_r}{4\pi}\right)$$

$$= P_t g_t g_r \left(\frac{\lambda^2}{4\pi R}\right)^2$$

$$= \frac{P_t g_t g_r \lambda^2}{16\pi^2 R^2}$$

$$\Rightarrow 10 \log_{10} P_r = 10 \log \left(\frac{P_t g_t g_r \lambda^2}{16\pi^2 R^2}\right)$$

$$= 10 \log (P_t g_t g_r \lambda^2) - 10 \log (4\pi R)^2$$

$$\Rightarrow P_{r_{dB}} = 10 \log (P_t g_t g_r \lambda^2) - 20 \log 4\pi R$$

$$\Rightarrow P_{r_{dB}} = 10 \log P_t + 10 \log g_t + 10 \log g_r + 20 \log \lambda - 20 \log (4\pi R)$$

$$= 10 \log P_t + 10 \log g_t + 10 \log g_r - (20 \log 4\pi R - 20 \log \lambda)$$

$$= 10 \log P_t + 10 \log g_t + 10 \log g_r - 20 \log \left(\frac{4\pi R}{\lambda}\right)$$

$$\therefore \boxed{P_{r_{dB}} = P_t + G_t + G_r - 20 \log \frac{4\pi R}{\lambda} \quad dBW}$$

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Plane wave Propagation in loseless Dielectric [need for calculation]

$\sigma = 0$, phase constant $\beta = \omega\sqrt{\mu\epsilon}$

phase velocity, $v_p = \frac{1}{\sqrt{\mu\epsilon}}$

Attenuation constant, $\alpha = 0$ (no loss)

Blossless dielectric
- no electrical conductivity
- $\sigma = 0$
- it only stores energy

$\therefore$ Intrinsic independence,

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_r \epsilon_0}} = \frac{377}{\sqrt{\epsilon_r}}$$

Lossy media are characterized by [$\sigma \neq 0$]

① Good conductor [$\sigma >> \omega\epsilon$]

② Poor conductor

③ lossy dielectric [$\sigma << \omega\epsilon$]

Good Conductor:

For a good conductor, it is required $\sigma >> \omega\epsilon$ or $\frac{\sigma}{\omega\epsilon} >> 1$

From complex frequency solution, the general propagation constant,

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \quad \text{——— (i)}$$

From (i),

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$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \quad \text{---(2)}$$

$$= \sqrt{j\omega\mu\left(1 - \frac{j\sigma}{\omega\epsilon}\right)j\omega\epsilon}$$

$$= \sqrt{j^2\omega^2\epsilon\mu\left(1 - \frac{j\sigma}{\omega\epsilon}\right)}$$

$$= j\omega\sqrt{\epsilon\mu\left(1 - \frac{j\sigma}{\omega\epsilon}\right)}$$

$$= j\omega\sqrt{\epsilon\mu\left(-\frac{j\sigma}{\omega\epsilon}\right)} \quad \left[\frac{\sigma}{\omega\epsilon} >> 1\right]$$

$$= j\omega\sqrt{\mu\epsilon}\sqrt{\frac{\sigma}{\omega\epsilon}}\sqrt{-j}$$

$$= j\sqrt{\omega\mu\epsilon}\sqrt{\frac{\sigma}{\epsilon}}\sqrt{-j} \quad \text{---(2)}$$

$$= j\sqrt{\omega\mu\sigma}\sqrt{-j} \quad \text{---(3)}$$

(i) $\sigma >> \omega\epsilon$

$\Rightarrow \frac{\sigma}{\omega\epsilon} >> 1$

$\Rightarrow 1 - \frac{j\sigma}{\omega\epsilon}$

$\Rightarrow \boxed{-\frac{j\sigma}{\omega\epsilon}}$

(2) $\sigma + j\omega\epsilon$

$= j\omega\epsilon\left(1 + \frac{\sigma}{j\omega\epsilon}\right)$

$= j\omega\epsilon\left(1 + \frac{j\sigma}{j^2\omega\epsilon}\right)$

$= \boxed{j\omega\epsilon\left(1 - \frac{j\sigma}{\omega\epsilon}\right)}$

$[j^2 = -1]$

From eq. (3), excluding constants,

$\gamma \propto \gamma_\omega = j\sqrt{\omega\mu\epsilon}\sqrt{j}$

$= j\sqrt{\omega\mu\epsilon}\left(\frac{1-j}{\sqrt{2}}\right)$ ---(3)

$= j\sqrt{\omega\mu\epsilon}\left(\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}\right)$

$= \sqrt{\omega\mu\epsilon}\left(\frac{j}{\sqrt{2}} - \frac{j^2}{\sqrt{2}}\right)$

(3) $\sqrt{-j} = \left(\frac{1-j}{\sqrt{2}}\right)$

$= \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$

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$$= \sqrt{\omega\mu\epsilon}\left(\frac{j}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)$$

$$= \sqrt{\omega\mu\epsilon}\left(\frac{1+j}{\sqrt{2}}\right)$$

$$= (1+j)\sqrt{\frac{\omega\mu\epsilon}{2}}$$

$$= (1+j)\sqrt{\frac{2\pi f\mu\epsilon}{2}}$$

$$= (1+j)\sqrt{\pi f\mu\epsilon}$$

This is for permitivity.

Again from (2),

$$\gamma = j\sqrt{\frac{\omega\mu\epsilon\sigma}{\epsilon}}\sqrt{-j}$$

$$= j\sqrt{\omega\mu\sigma}\sqrt{-j}$$

$$= j\sqrt{\omega\mu\sigma}\left(\frac{1-j}{\sqrt{2}}\right)$$

$$= \sqrt{\omega\mu\sigma}\ \frac{1+j}{\sqrt{2}}$$

$$= \sqrt{\frac{\omega\mu\sigma}{2}}\ (1+j)$$

$$= \sqrt{\frac{2\pi f\mu\sigma}{2}}\ (1+j)$$

$$= \sqrt{\pi f\mu\sigma}\ (1+j)$$

This is for conduction.

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TOPIC NAME : Lecture 7

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Tranemission lines

Mid → lec 6
CT-2 → 5, 6

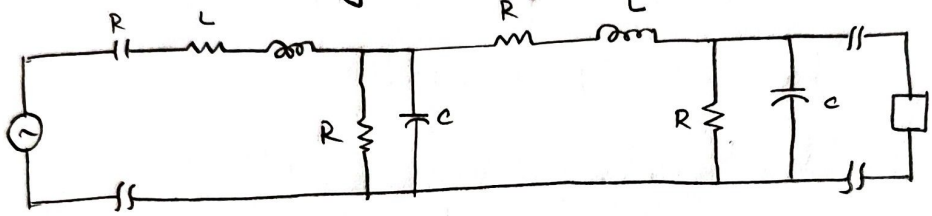
- If irregular arrangement happens → not distributed linearly
- Distribution depends on → Induction, resistance, capacitance
- From kirchoff's voltage law,

$\lambda = c/f$

$v = f\lambda$

Inductance → e^- flow

Capacitance → charge stored



$$v_s(z,t) = \frac{i(z,t)\Delta z}{\Delta z} + \frac{L\Delta z}{\Delta z}\frac{\partial i(z,t)}{\partial t} + v(z,t) + \frac{\frac{\partial v(z,t)}{dz}\Delta z}{\Delta z}$$

$$\Rightarrow 0 = Ri(z,t) + L\frac{\partial i(z,t)}{\partial t} + \frac{\partial v(z,t)}{\partial z} \quad \text{—①}$$

$$\Rightarrow -\frac{\partial v}{dt} = Ri + L\frac{\partial i}{\partial t} \quad \text{—②}$$

$$-\frac{\partial v}{\partial z} = Ri + L\frac{\partial i}{\partial t}$$

Differentiating w.r.t z, we get,

$$-\frac{\partial^2 v}{\partial z^2} = R\cdot\frac{\partial i}{\partial z} + L\frac{\partial^2 i}{\partial t\,\partial z}$$

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From kirchoffic current law,

$$-\frac{\partial i}{\partial z} = GV + C\frac{\partial v}{\partial t}$$

$$\Rightarrow -\frac{\partial^2 i}{\partial z \partial t} = G\frac{\partial v}{\partial t} + C\frac{\partial^2 v}{\partial t^2}$$

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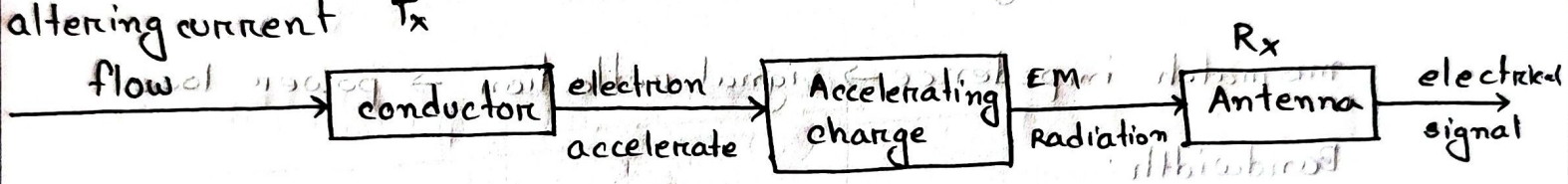
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~~What scares me?~~

Types of Antennas

Antenna is a communication device that radiates or receive eletricalmagnetic waves into free space.



frequency → number of cycles per seconds

wavelength → Distance between two wave peak

$$\lambda = \frac{c}{f}$$

Radiation pattern

① Isotropic → equal radiation in each direction (ideal + x practical used as reference)

② Omnidirectional → equal direction horizontally (circular) 360° routers

③ Directional → Focus beam, high gain, directivity

Directivity : Ability to focus energy to specific direction

$$D = \frac{\text{Power density for given direction}}{\text{Avg Power density}}$$

Gain : Measure of antenna's ability to direct & amplify signal

$$G = \eta \times D$$

= efficiency x directivity

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Efficiency: Ratio of radiated pattern

$$\eta = \frac{P_{out} / P_{radiation}}{P_{input}}$$

Impedence: Ratio of voltage to current at any antenna input

$$Z = R + jX$$

mismatch impedence → signal reflection → power loss

Bandwidth:

Type 2

By physical structure

1. Wire → simple & common, → Dipole
   - ↳ Radio + TV
   - → Monopole
2. Aperture → use an openning to radiate waves
   - ↳ ~~parabolic reflector~~ Radar
3. Reflector → use reflecting surface to direct
   - ↳ satellite, space
4. Array → combo of multiple, ↑ directivity, steering beam
   - ↳ phased array
5. Microstrip (Patch) → Flat compact, printed on circuit board
   - ↳ mobile, GPS
6. lens → use dielectric lens to focus
   - ↳ high freq. system

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